

```
$ boost
```

MLFQ BOOST TEST: Demotion & Promotion Cycles

```
*****
```

[TEST 1] Initial Priority Check

```
@ T0 | Initial | PID:3 Q0 slices:0/3
```

```
>> PASS: New process starts at Q0 (highest)
```

[TEST 2] Demotion Test (CPU-bound work)

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```
Running CPU-intensive work to trigger demotion...
```

Queue	Quantum	Description
Q0	3 ticks	Highest (interactive)
Q1	6 ticks	High
Q2	12 ticks	Medium
Q3	24 ticks	Low (CPU-bound)

(1 tick ~ 100ms; demotion at slices >= quantum)

Monitoring slices and demotions:

```
[T1] Q0: slices 0->1 (need 3)
```

```
[T2] Q0: slices 1->2 (need 3)
```

```
[T3] DEMOTE: Q0 --> Q1 (used 3/3)
```

```
[T4] Q1: slices 0->1 (need 6)
```

```
[T5] Q1: slices 1->2 (need 6)
```

```
[T6] Q1: slices 2->3 (need 6)
```

```
[T7] Q1: slices 3->4 (need 6)
```

```
[T8] Q1: slices 4->5 (need 6)
```

```
[T9] DEMOTE: Q1 --> Q2 (used 6/6)
```

```
[T10] Q2: slices 0->1 (need 12)
```

```
[T11] Q2: slices 1->2 (need 12)
```

```
@ T11 | After work | PID:3 Q2 slices:2/12
```

Demotion Summary:

```
* Q0->Q1 @ tick 3 (expected: 3 ticks in Q0)
```

```
* Q1->Q2 @ tick 9 (expected: 6 more in Q1)
```

```
>> PASS: Demoted from Q0 to Q2
```

[TEST 3] Manual Boost (boostproc syscall)

```
@ T12 | Before boost | PID:3 Q2 slices:2/12
```

```
Invoking boostproc()...
```

```
@ T12 | After boost | PID:3 Q0 slices:0/3
```

```
Boost ran between T12 and T12
```

```
>> PASS: Boosted to Q0, slices reset
```

[TEST 4] Re-demotion After Boost

```
Verifying demotion still works post-boost...
```

```
(Auto-boost may interfere if BOOST_INTERVAL is short)
```

```
[T15] DEMOTE: Q0 --> Q1
```

```
@ T15 | Final | PID:3 Q1 slices:0/6
```

```
>> PASS: Demotion works after boost
```

[TEST 5] Automatic Periodic Boost

```
Waiting for automatic boost to trigger...
```

```
(Tests starvation prevention)
```

[T21] DEMOTE: Q1 --> Q2
[T33] DEMOTE: Q2 --> Q3
[T79] AUTO BOOST: Q3 --> Q0

>> PASS: Auto periodic boost working

TEST SUMMARY

Runtime: 79 ticks (~7.9 sec)

Initial Queue: Q0

Max Demotion: Q2

Manual Boost: OK

Auto Boost: Detected

>>> ALL CORE TESTS PASSED <<<

\$

xv6 kernel is booting

init: starting sh

\$ cpubnd

CPU-BOUND WORKLOAD TEST

Running heavy CPU computation for ~30 sec

Observing priority demotion: Q0 -> Q1 -> Q2 -> Q3

[INIT] PID: 3 | Queue: Q0 | Slices: 0

Executing intensive loop (pure CPU, no I/O)...

Demotion schedule:

- * Q0: 0-2 ticks (start)
- * Q1: 2-6 ticks (1st demotion)
- * Q2: 6-14 ticks (2nd demotion)
- * Q3: 14+ ticks (final level)

[%02d] Queue=Q0, Slices=0
[%02d] Queue=Q1, Slices=0
[%02d] Queue=Q2, Slices=1
[%02d] Queue=Q3, Slices=1
[%02d] Queue=Q4, Slices=1
[%02d] Queue=Q5, Slices=1
[%02d] Queue=Q6, Slices=1
[%02d] Queue=Q7, Slices=1
[%02d] Queue=Q8, Slices=1
[%02d] Queue=Q9, Slices=1
[%02d] Queue=Q10, Slices=2
[%02d] Queue=Q11, Slices=2
[%02d] Queue=Q12, Slices=2
[%02d] Queue=Q13, Slices=2
[%02d] Queue=Q14, Slices=2
[%02d] Queue=Q15, Slices=2
[%02d] Queue=Q16, Slices=2
[%02d] Queue=Q17, Slices=2
[%02d] Queue=Q18, Slices=0
[%02d] Queue=Q19, Slices=0
[%02d] Queue=Q20, Slices=0
[%02d] Queue=Q21, Slices=0
[%02d] Queue=Q22, Slices=1
[%02d] Queue=Q23, Slices=1
[%02d] Queue=Q24, Slices=1
[%02d] Queue=Q25, Slices=1
[%02d] Queue=Q26, Slices=1
[%02d] Queue=Q27, Slices=1
[%02d] Queue=Q28, Slices=1
[%02d] Queue=Q29, Slices=2
[%02d] Queue=Q30, Slices=2

... continuing to completion ...

[%02d] Queue=Q31, Slices=2

... continuing to completion ...

[%02d] Queue=Q31, Slices=2
[%02d] Queue=Q32, Slices=2
[%02d] Queue=Q33, Slices=2
[%02d] Queue=Q34, Slices=2
[%02d] Queue=Q35, Slices=2
[%02d] Queue=Q36, Slices=2
[%02d] Queue=Q37, Slices=2
[%02d] Queue=Q38, Slices=2
[%02d] Queue=Q39, Slices=2
[%02d] Queue=Q40, Slices=2
[%02d] Queue=Q41, Slices=2
[%02d] Queue=Q42, Slices=2
[%02d] Queue=Q43, Slices=3
[%02d] Queue=Q44, Slices=3
[%02d] Queue=Q45, Slices=3
[%02d] Queue=Q46, Slices=3
[%02d] Queue=Q47, Slices=3
[%02d] Queue=Q48, Slices=3
[%02d] Queue=Q49, Slices=3

RESULT: Queue=Q3 | Total Slices=5
STATUS: [PASS] Reached lowest queue Q3

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```
$ iobnd
```

```
=== Comprehensive I/O-Bound Test ===
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```
This process will alternate between brief work and sleep.
```

```
It should STAY in Q0 (never demote) because it yields frequently.
```

```
Starting: PID=4, Priority=Q0, TimeSlices=0
```

```
Running 30 iterations of: brief_work() -> sleep(1_tick)
```

```
Expected: Priority stays Q0 throughout (slices reset after sleep)
```

```
Iteration 0: Priority=Q0, TimeSlices=0
```

```
Iteration 5: Priority=Q0, TimeSlices=0
```

```
Iteration 10: Priority=Q0, TimeSlices=0
```

```
  [Checkpoint] Still at Q0
```

```
Iteration 15: Priority=Q0, TimeSlices=0
```

```
Iteration 20: Priority=Q0, TimeSlices=0
```

```
  [Checkpoint] Still at Q0
```

```
Iteration 25: Priority=Q0, TimeSlices=0
```

```
  [Checkpoint] Still at Q0
```

```
Final Result: Priority=Q0, TimeSlices=0
```

```
✓ SUCCESS: Stayed in Q0 (high priority maintained!)
```

```
  Demonstrates I/O-bound processes get preferential treatment.
```

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```

```
$ testmlfq
=== Comprehensive Mixed Workload Test ===
Running CPU-bound and I/O-bound processes concurrently.
This shows the scheduler's fairness: CPU-bound demotes, I/O-bound stays high.
```

```
[CPU-BOUND] Starting long computation...
[CPU-BOUND] Iteration 0: Q0 (slices=1)
[I/O-BOUND] Starting brief work + frequent sleeps...
[I/O-BOUND] Iteration 0: Q0 (slices=0)
[I/O-BOUND] Iteration 5: Q0 (slices=0)
[CPU-BOUND] Iteration 10: Q2 (slices=1)
[I/O-BOUND] Iteration 10: Q0 (slices=0)
[CPU-BOUND] Iteration 20: Q1 (slices=3)
[I/O-BOUND] Iteration 15: Q0 (slices=0)
[I/O-BOUND] Iteration 20: Q0 (slices=0)
[CPU-BOUND] Iteration 30: Q2 (slices=5)
[I/O-BOUND] Final: Q0 (TimeSlices=0)
[CPU-BOUND] Final: Q3 (TimeSlices=1)
```

```
=== Test Complete ===
```

```
Expected Behavior:
```

```
  CPU-bound:   Demoted to Q2 or Q3 (lower priority)
```

```
  I/O-bound:   Stayed in Q0 or Q1 (higher priority)
```

```
  Result:      I/O-bound process got preference despite CPU competition
```

```
$
```

\$ diag

=== MLFQ Diagnostics ===

Initial state:

PID: 8

State: 4

Priority: Q0

Time Slices: 0

Running work bursts and checking time_slices:

Iteration %2d: slices=0 priority=Q0

Iteration %2d: slices=1 priority=Q0

Iteration %2d: slices=2 priority=Q0

Iteration %2d: slices=3 priority=Q0

Iteration %2d: slices=4 priority=Q1

Iteration %2d: slices=5 priority=Q1

Iteration %2d: slices=6 priority=Q1

Iteration %2d: slices=7 priority=Q1

Iteration %2d: slices=8 priority=Q1

Iteration %2d: slices=9 priority=Q2

Iteration %2d: slices=10 priority=Q2

Iteration %2d: slices=11 priority=Q2

Iteration %2d: slices=12 priority=Q2

Iteration %2d: slices=13 priority=Q2

Iteration %2d: slices=14 priority=Q2

Iteration %2d: slices=15 priority=Q0

*** GOOD: Process demoted to Q1 after 0 slices! ***

Final state:

Priority: Q1

Time Slices: 0

=== DIAGNOSIS ===

Problem: time_slices never incremented

Possible causes:

1. Timer interrupts not firing (check clockintr)
2. which_dev != 2 in usertrap()
3. p->time_slices++ not executing

Add debug prints in kernel/trap.c:

```
printf("Timer: which_dev=%d pid=%d\n", which_dev, p->pid);
```

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\$ purecpu

=== Pure CPU-Bound Test ===

Starting: PID=10, Q0, slices=0

Running continuous CPU work (no interruptions)...
This will take several seconds...

Work complete! Checking final state...

Final: PID=10, Q2, slices=5

Expected behavior:

- Should have used many timer ticks
- Should have demoted through queues
- Should be at Q3 (or Q2/Q3)

✓ SUCCESS: Demoted to Q2

\$

\$ fair

=== Long-Running Fairness Test (Week 3) ===

Running 3 different workloads for ~200 ticks

1. Pure CPU-bound (should demote but get boosted)
2. Pure I/O-bound (should stay high priority)
3. Mixed workload

[CPU-BOUND] Starting...

[CPU-BOUND] Tick 9690: Q0 (slices=0)

[Parent] All processes running...

[I/O-BOUND] Starting...

[MIXED] Starting...

[I/O-BOUND] Tick 9696: Q0 (slices=0)

[MIXED] Tick 9698: Q0 (slices=1)

[CPU-BOUND] BOOST #1 detected at tick 9700 (Q1->Q0)

[CPU-BOUND] Tick 9709: Q1 (slices=4)

[CPU-BOUND] Tick 9728: Q3 (slices=0)

[CPU-BOUND] Tick 9748: Q3 (slices=15)

[I/O-BOUND] Tick 9749: Q0 (slices=0)

[MIXED] Tick 9759: Q2 (slices=5)

[CPU-BOUND] FINAL: Q3, 1 boosts detected

[CPU-BOUND] ✓ Received at least one boost!

[I/O-BOUND] Tick 9799: Q0 (slices=0)

[MIXED] Tick 9819: Q1 (slices=1)

[I/O-BOUND] Tick 9849: Q0 (slices=0)

[MIXED] Tick 9879: Q2 (slices=10)

[MIXED] FINAL: Q3

[I/O-BOUND] FINAL: Q0

[I/O-BOUND] ✓ Maintained high priority

=== Fairness Test Complete ===

All three workloads completed successfully!

Expected results:

- CPU-bound: Experienced boosts (no starvation)
- I/O-bound: Stayed in high priority queues
- Mixed: Balanced between Q1-Q2

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```
$ starve
=== Starvation Prevention Test (Week 3) ===
Testing automatic priority boosting every 100 ticks
Running 3 CPU-bound processes concurrently
All processes should get CPU time due to periodic boosting
```

```
[Process 0] Started (PID=17)
[Process 0] Tick 10627: Q0 (slices=0)

[Parent] Waiting for all processes to complete...
[Process 1] Started (PID=18)
[Process 1] Tick 10630: Q0 (slices=0)
[Process 2] Started (PID=19)
[Process 2] Tick 10633: Q0 (slices=0)
[Process 0] Tick 10639: Q1 (slices=3)
[Process 1] Tick 10645: Q1 (slices=3)
[Process 2] Tick 10651: Q1 (slices=3)
[Process 0] Tick 10657: Q2 (slices=3)
[Process 0] Tick 10664: Q2 (slices=9)
[Process 1] Tick 10670: Q2 (slices=3)
[Process 1] Tick 10676: Q2 (slices=9)
[Process 2] Tick 10682: Q2 (slices=3)
[Process 2] Tick 10688: Q2 (slices=9)
[Process 0] COMPLETED at tick 10694: Final Q3
[Parent] Process 0 finished
[Process 1] COMPLETED at tick 10697: Final Q3
[Parent] Process 1 finished
[Process 2] COMPLETED at tick 10701: Final Q0
[Parent] Process 2 finished
```

```
=== Starvation Test Complete ===
```

```
Analysis:
```

- All processes should have completed
- Processes should have been boosted to Q0 every ~100 ticks
- Even low-priority processes got CPU time
- No process was starved!

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